Absorbing phase transition in a queueing model of coupled adaptive agents

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What decides whether people do things together or separately? Many activities cannot be carried out alone, and an individual must rank them against the private tasks competing for the same time. We address this within the priority-queue description of human activity by letting each agent *choose* the priority of a shared task rather than drawing it from a fixed distribution: the value of the joint activity, discounted by the estimated risk that the partner will not take part. Participation becomes strategic, and the model acquires a phase transition. A coupled phase, in which joint activity is sustained, is separated from an absorbing solitary phase by a saddle-node bifurcation that we obtain in closed form. The transition is discontinuous and the solitary phase is absorbing, so collapse is irreversible unless an agent persists unilaterally for of order one memory time, a cost we also compute. The heavy-tailed interevent statistics that motivate queueing models of human dynamics survive only in a narrow window at the transition, and there the exponent is fixed by the fraction of time spent coupled rather than by queue length: the universality classes of the non-strategic model do not survive the introduction of choice. On a network, attention divides as $1/(k+a)$ and fixes a critical degree beyond which no coupled state exists, so the solitary phase percolates according to the Molloy–Reed criterion with the second moment truncated at that degree — formally an attack on hubs, with no attacker. For group activities the critical degree falls as the $(m-1)$th root, implying a maximum group size. The pair results are analytic and reproduced by simulation; the network and group thresholds are mean-field predictions, tested against simulations that locate the transition and characterise the fluctuations that displace it, but that do not resolve its critical behaviour. The transition organises two quantities already measured in communication records: a finite capacity for keeping ties active, and the decay of ties whose rhythm is interrupted.

I. INTRODUCTION

Human dynamics asks how people allocate their time among the things they might do, and what the timing of the resulting activity reveals about the decision process behind it. Its standard framework is queueing theory [1–3]: an individual's to-do list is a priority queue executed highest-priority-first, which produces power-law interevent time distributions $P(\tau) \sim \tau^{-\alpha}$, with $\alpha = 1$ and $\alpha = 3/2$ identified as universality classes and corroborated by electronic and postal correspondence data [1, 3, 4], the $\alpha = 3/2$ class going back to the single-server analysis of Cobham [5].

The question we address is one this framework leaves open. A person's list is not made only of things that can be done alone. A meeting, a conversation, a shared meal happen only if somebody else chooses them at the same moment. How should such an activity be ranked against the private tasks competing for the same hour, and what becomes of the timing statistics when everyone ranks that way?

The first half of the question has been posed. A minimal two-agent extension [6] gives each agent one *interacting* task I, executable only when both agents select it in the same step, and one aggregate *non-interacting* task O standing for $L_j - 1$ private tasks, predicting a countable family of exponents, $\alpha = 1 + 1/\max_j(L_j - 1)$, interpolating between 2 and 1: the more distracted agent sets the exponent.

In that treatment the priority of the interacting task is drawn independently from a fixed density. This is the natural assumption if the shared activity is exogenous, but it removes the feature that makes joint tasks distinctive: an agent who is repeatedly stood up should stop offering. Here we make the interacting-task priority a decision variable. The agents estimate each other's participation rate from observation and set their priority accordingly. This changes the object from a stochastic process into a coupled inference problem and, as we show, changes its phenomenology substantially. Letting a strategic variable respond to the state it helps to produce is the defining move of coevolutionary games, in which strategies, payoffs, interaction network or reputation adapt alongside one another and typically generate cooperative states the static game does not possess [10]. What coevolves here is narrower: the priority a task is given inside a queue, and the estimate of somebody else's behaviour that sets it.

Once participation is chosen rather than imposed, the object of study changes. The question is no longer the shape of one distribution but the collective state of a set of agents, and the natural instrument for it is statistical mechanics. Human dynamics already borrows from that apparatus at the level of exponents and universality classes, and interaction is known from the theory of critical phenomena to be decisive for universality [7]. The statistical-physics treatment of human activity, com-

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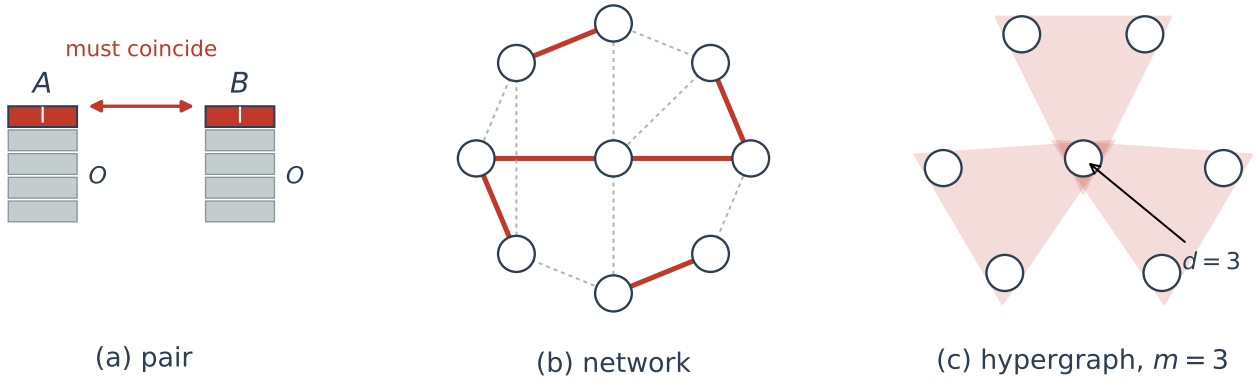


FIG. 1. The three settings. An agent holds one interacting task per shared activity plus an aggregate private task O standing for $L-1$ individual tasks, and executes one item per step. (a) A pair: I executes only if both select it. (b) A network: one interacting task per neighbour, so attention is divided; solid links are coupled relations, dashed ones have collapsed. (c) A hypergraph: an activity of size m executes only if all m members select it, and an agent may belong to d groups.

munication, collective behaviour and social interaction patterns is by now a field in its own right, reviewed in Ref. [9], and the present model sits inside it. We argue that here the informative structure lies one level up. The model has two phases, an order parameter that separates them, and a transition of definite character; the exponents turn out to be a property of where the system sits relative to that transition rather than the object of study. Phases and transitions, not exponents, are what the model has to say about human activity.

Strategic queueing is of course an established subject, from Naor [8] onward, but it concerns customers deciding whether to join, balk, or purchase priority in a service queue. What is at issue here is different: the priority of a task that cannot be executed unilaterally, so that the queueing model is simultaneously a coordination game with an absorbing failure state. The resulting phenomenology connects to two literatures with little contact so far with the queueing account of human dynamics. The capacity constraint derived below has been measured directly: mobile-phone records show a finite communication capacity limiting how many ties an individual holds active, with larger networks maintaining weaker ties rather than more communication [11–13]. So has the timescale condition: ties interrupted for much longer than their established rhythm decay [14]. And the percolation criterion we obtain is formally that of intentional attack [15–17], with no adversary present — attention scarcity removes high-degree relations by itself.

Here we investigate models of human dynamics when the priority of the shared task becomes a choice rather than a draw. The paper is laid out as follows. Section II defines the model, derives the policy that replaces the fixed priority density, and reads its parameters as statements about behaviour. Section III solves the symmetric pair and gives the phase boundary in closed form. Section IV states the simulation protocol and classifies every result as exact, mean-field or measured. Section V reports the pair: the loss of the universality classes, the bimodality and irreversibility of the outcome, the statistics of escape, the phase boundary in c and L, and the price of recovery. Section VI puts the agents on a network, where attention divides as $1/(k+a)$ and the solitary phase percolates, and Sec. VII on a hypergraph, where an activity requires all of its members and a maximum group size follows. Section VIII collects what the transition organises.

II. MODEL

A. Definition

We state the model in full, so that it can serve as a reference point. It retains the substrate of Ref. [6] and adds rule (vi).

(i) *Tasks.* Agent j holds one aggregate private task O standing for $a_j \equiv L_j - 1$ individual tasks, and one interacting task I for each shared activity it belongs to.

(ii) *Priorities.* The private-task priority is the largest of a_j independent uniform variates, of density $f_{O_j}(x) = a_j x^{a_j-1}$ on $[0,1]$, redrawn whenever O is executed. The interacting-task priority is x_{I_j}, held until the task is acted on.

(iii) *Selection.* Each agent executes the single highest-priority item in its list per step.

(iv) *Execution.* A shared activity executes only if every one of its members selects it in the same step.

(v) *Payoffs.* Each participant receives R_I when the activity executes; an agent whose offer is not reciprocated receives $x_{O_j} - c$, the step being wasted; otherwise an agent receives x_{O_j}, the realised value of its private task.

(vi) *Policy.* x_{I_j} is set by Eq. (2) from an estimate $\hat{p}$ of the probability that the others participate, maintained as an exponential moving average with retention $\lambda = e^{-1/\tau_{\mathrm{mem}}}$ over observed selections. Independently, an agent offers regardless of belief with probability ϵ per step.

Rules (i)–(v) are those of Ref. [6], in which x_{I_j} is drawn from a fixed density; rule (vi) replaces that draw by a decision and is the whole of what we add. The pair, network and hypergraph settings of Fig. 1 differ only in how many interacting tasks an agent holds and how many members an activity requires. Two amendments are needed once an agent holds more than one interacting task, and are given in Sec. VI.

The exploration rate ϵ in (vi) is not a numerical convenience. At $\epsilon = 0$ the state $p = 0$ is strictly absorbing: an agent that stops participating never observes its partner participating and therefore never revises. Escape requires two agents to explore in the same step, at rate ϵ^2.

B. Priority as a decision

Since O is drawn afresh each step, the probability that j selects I is

$$p_j = \Pr\big(x_{O_j} < x_{I_j}\big) = x_{I_j}^{a_j}, \tag{1}$$

so the agent controls a propensity, not a decision: with a_j large, even a high x_{I_j} yields little participation.

The cost c of an unreciprocated offer is what makes a model of the partner necessary; at $c = 0$ the optimal policy is $x_{I_j} = 1$ whatever the partner does.

Writing $\hat{p}$ for the agent's estimate of its partner's participation probability, selecting I is preferable whenever $\hat{p}R_I + (1-\hat{p})(x_{O_j} - c) > x_{O_j}$, giving the optimal priority

$$x^*_{I_j} = \min\left\{1,\ \max\left\{0,\ R_I - c\,\frac{1-\hat{p}}{\hat{p}}\right\}\right\}, \tag{2}$$

clipped to the unit interval: *the value of the interaction discounted by the risk that the partner will not participate.* Estimates are maintained by an exponential moving average with retention $\lambda = e^{-1/\tau_{\mathrm{mem}}}$ over the partner's observed selections.

C. Behavioural reading of the assumptions

Four elements of rule (vi) and of Eq. (2) carry the conclusions below, so it is worth stating what each asserts about behaviour.

The cost c of an unreciprocated offer is the price of turning up and finding nobody there: the hour is spent, the private task that would have filled it is not done, and no joint activity takes place. It is what makes a model of the partner necessary rather than ornamental, since at $c = 0$ offering is free, $x_{I_j} = 1$ is optimal whatever the partner does, and the absorbing state disappears along with the incentive to infer. Read empirically, c is the asymmetry between a missed appointment and a quiet evening, and the model's claim is that this asymmetry, rather than the value of the activity, is what makes joint activity precarious.

The memory time τ_{mem} is the horizon over which a relation is judged: a short one describes a partner assessed on the last few encounters, a long one a partner assessed on a long history. It fixes the variance of the estimate, of order $1/\tau_{\mathrm{mem}}$ for an exponential moving average, and hence the rate at which fluctuations carry a coupled pair across the unstable fixed point. It is also the natural unit for the price of recovery, since what a returning agent must overcome is an accumulated record of not interacting.

The exploration rate ϵ is the probability of offering with no expectation of reciprocation — calling somebody one has not heard from, proposing something that will probably be declined. It is the only route back, because an agent that offers only when its own estimate recommends it never acquires evidence against that estimate. Its role is structural rather than quantitative: the intermittent window of Sec. V D exists for $\epsilon > 0$ and not at all at $\epsilon = 0$.

Holding rather than decaying beliefs during silence is the difference between reading no news as no information and reading it as refusal. An agent that decays its estimate through an interval in which nothing could have been observed treats absence of evidence as evidence of absence; one that holds it maintains a representation of its partner with no input at all. These are distinct attitudes to a quiet period, and Sec. V C shows that which one is adopted decides whether a rarely repeated joint activity is viable.

D. Levels of modelling

We distinguish three policies by what each agent represents.

Level 0: No partner model: $x_I = R_I$, clipped.

Level 1: Partner model: $\hat{p}$ tracked, Eq. (2) applied.

Level 2: Self model: the agent additionally tracks $\hat{s}$, its estimate of *the partner's estimate of itself*, and evaluates a finite-horizon rollout of the coupled dynamics in which its own participation raises $\hat{s}$, which raises the partner's priority through Eq. (2), which raises the partner's participation. It commits to

$x_I = 1$ when that rollout dominates the myopic choice.

Level 2 is the minimal policy requiring an agent to represent itself as an object in another's model; the ladder is that of the cognitive-hierarchy and level-k literature [18].

III. FIXED POINTS

For a symmetric pair the participation probability obeys $p \mapsto \mathcal{F}(p)$ with

$$\mathcal{F}(p) = \Big(\min\big\{1, \max\{0,\, R_I - c(1-p)/p\}\big\}\Big)^a. \qquad (3)$$

Setting $\mathcal{F}(p) = p$ and requiring tangency gives the saddle-node in closed form,

$$p^* = (ac)^{a/(a+1)}, \qquad R_c = (p^*)^{1/a} + \frac{c}{p^*} - c. \qquad (4)$$

For $c = 0.35$ this gives $R_c = 0.8332$ at $p^* = 0.5916$ for $L = 2$, and $R_c = 0.9818$ at $p^* = 0.7885$ for $L = 3$.

Equation (4) is meaningful only while $p^* \le 1$, that is

$$a \le 1/c. \qquad (5)$$

Above this bound the tangency lies outside the accessible range and there is no interior bifurcation: the coupled phase is created discontinuously at the clipping boundary $x_I = 1$. At $c = 0.35$ this restricts the continuous transition to $L \le 3.86$. Figure 2 shows the return map and the critical line.

Below R_c only the solitary fixed point $p = 0$ exists. Above it there are two stable phases, coupled and solitary, separated by an unstable point whose position rises with a: more private tasks enlarge the basin of solitude. The structure is that of a stag hunt [19], whose behaviour on growing and structured networks has been studied in its own right [20], with the solitary phase risk-dominant and absorbing at $\epsilon = 0$.

IV. SIMULATION PROTOCOL AND STATUS OF THE RESULTS

This section states the numerical procedure in full and then labels every quantitative claim in the paper by the kind of evidence behind it. All the code is at the repository given in the acknowledgments, and each result below names the script that produces it.

A. Protocol

Update. Time is discrete and agents are updated in parallel, so their order is immaterial. At each step every agent draws its private-task priority as $x_{O_j} = u^{1/a_j}$ with u uniform on $[0, 1]$, realising the density f_{O_j} of rule (ii); sets the priority of each interacting task by rule (vi); and selects its single highest-priority item. A shared activity executes if all of its members selected it, and beliefs are then updated by the exponential moving average of rule (vi) from what each agent observed. A private-task priority is redrawn every step, because O is executed whenever I is not selected; an interacting-task priority is held until the task is acted on, which on a network means until an offer is made (Sec. VI).

Initial conditions. Unless stated otherwise all beliefs start at $\hat{p} = \hat{s} = 0.85$, inside the basin of the coupled state, and x_I starts at R_I. Runs therefore begin coupled, and a collapse is an escape from the coupled state rather than a failure to reach it. Because the transition is discontinuous and the solitary state absorbing, the initial condition matters and we report its effect explicitly, for the pair in Table I and for the network in Table II of the Supplemental Material [21].

Length and burn-in. Pair runs are 2×10^5 steps with the first 20% discarded, except where a different length is the point of the measurement: 1.2×10^6 steps for the tail indices of Table III, which need four decades of interevent times, and the ladder 2.5×10^4 to 8×10^5 steps in Supplemental Table I. Network runs are 4×10^5 steps with the first half discarded; the shorter runs used to establish that this is long enough are reported separately in Supplemental Table II. Hypergraph runs are 4×10^4 steps (6×10^4 for the memory-time scan) with the first half discarded.

Realisations. A realisation is one run with an independent random seed, which independently randomises the private-task priorities, the interacting-task draws, the exploration events and, on a network or hypergraph, the graph itself, since the growth rule draws from the same seeded stream. Every number reported as $x \pm y$ is the mean over realisations and the standard error of that mean, $y = \sigma/\sqrt{N}$, with N stated in every caption; tables use the compact form $x(e)$, where e is that error in units of the last digit of x, so $0.843(32)$ means 0.843 ± 0.032. Near the transition the distribution over realisations is bimodal and a mean is uninformative; there we report the fraction in each mode, and the spread rather than the standard error.

Classifying the phase. For a pair we use the occupancy ϕ: the fraction of post-burn-in steps on which the mean participation propensity $(p_A + p_B)/2$ exceeds $p^*/2$, with p^* the saddle-node value of Eq. (4). It is bounded in $[0, 1]$, the cut sitting midway between the absorbing state at $p = 0$ and the participation at which the coupled branch is born, and a realisation counts as coupled when $\phi > 0.75$, solitary when $\phi < 0.25$. Two windows are used, and the difference matters. Where ϕ is a time average it is taken over the whole post-burn-in trajectory; where the question is which state a realisation *ended* in — the bimodality of Sec. V E and the recovery experiments after it — it is taken over the final 20%, which turns it into a classifier of the end state and is why almost no realisation there lands between the two modes. The window is stated

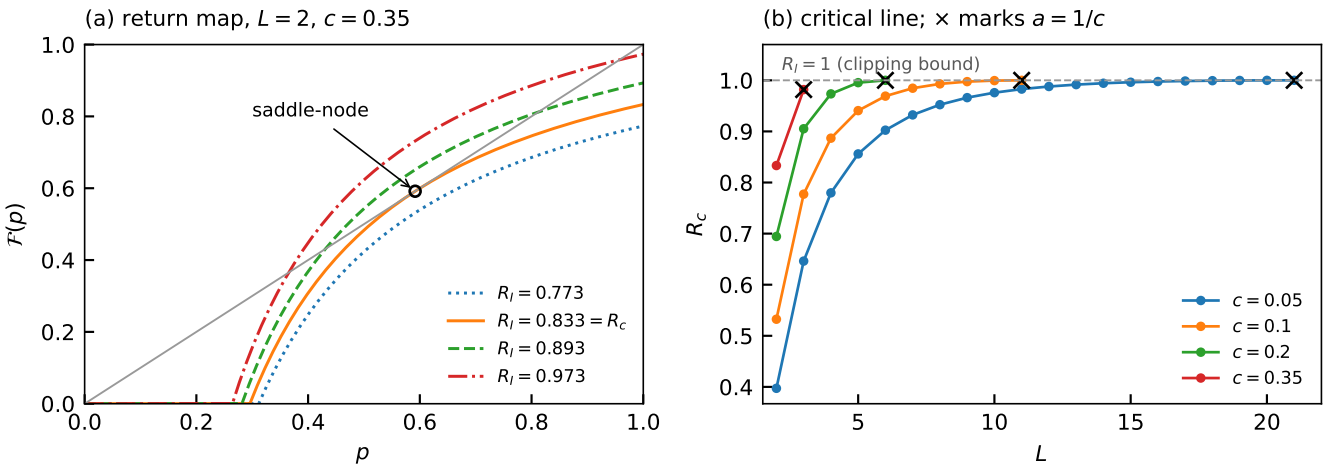


FIG. 2. (a) Return map $\mathcal{F}(p)$ for $L = 2$, $c = 0.35$, with the diagonal in grey; the two nontrivial fixed points are born in a saddle-node at $R_c = 0.8332$, $p^* = 0.5916$ (circle). (b) Critical line $R_c(L)$ from Eq. (4); crosses mark the validity boundary $a = 1/c$, Eq. (5).

with each measurement. For a network we classify each edge rather than each run: an edge is coupled when its time-averaged interaction rate w exceeds $10\epsilon^2$, ten times the rate at which two independently exploring endpoints coincide, which is the only activity a fully collapsed edge can have. We write f_c for the fraction of coupled edges and take as order parameter S, the largest connected component of the coupled subgraph as a fraction of n. Both cuts are tested in Sec. VI A, which also shows the distribution of per-edge rates the network cut is taken from.

Supplemental Material. The two widest parameter scans — the sensitivity of the pair and of the network to each parameter in turn — are reference tables rather than arguments, and are given in full as Tables I and II of the Supplemental Material [21]. The text quotes only the entries the argument uses.

B. Status of the results

The paper mixes three kinds of statement, and conflating them is easy because they appear in the same equations. *Exact* results follow from rules (i)–(vi) and from the algebra of the deterministic map those rules define, with no approximation; they are as good as the model. *Mean-field* results replace a fluctuating belief by its mean, or a graph by a locally tree-like one, or both; they are predictions whose accuracy has to be measured. *Numerical* results are measurements, with the number of realisations stated. Table I assigns each result to one of the three. In short: the pair phase boundary is exact and confirmed, the network and group thresholds are mean-field and quantitatively off in a direction we can explain and measure, and the transition is located but its critical behaviour is not characterised.

V. RESULTS

Parameters and protocol are those of Sec. IV: symmetric agents, $\tau_{\mathrm{mem}} = 200$ and 2×10^5 steps after a 20% burn-in unless stated otherwise, and the phase quantified by the occupancy ϕ.

A. Validation

Reproducing Ref. [6] with i.i.d. priorities recovers $\alpha = 1 + 1/\max_j(L_j - 1)$. Asymmetric pairs sit on the prediction, $(L_A, L_B) = (2, 3)$ giving $\alpha = 1.500$ and $(2, 5)$ giving 1.250, both stable across four decades of the tail; symmetric pairs approach from below ($L = 2$ gives 1.82 at the 90th percentile, drifting to 1.94 deeper in), as the logarithmic correction at $a_A = a_B$ requires. These are Hill estimates on the top 5% of 4×10^6 interevent times drawn from the exact coarse-grained model of Ref. [6]; over eight independent realisations the estimates are 1.4959(16) for $(2, 3)$ and 1.2500(8) for $(2, 5)$, so the agreement with the prediction is far outside the statistical error. The mean interevent time diverges with the observation window as $T^{2-\alpha}$, so for $\alpha \leq 2$ the interaction has no characteristic timescale.

B. Modelling the partner's model

Level-2 agents remain coupled well below the level-1 threshold. Over 40 independent realisations per point at $\tau_{\mathrm{mem}} = 50$ and $\epsilon = 0$, the fraction of realisations ending coupled at $L = 3$ is 0.00 for level-1 pairs at every $R_I \leq 0.98$ and 0.05 at $R_I = 0.99$, rising to 0.97 only at $R_I = 1.00$, whereas level-2 pairs end coupled in every realisation at every $R_I \geq 0.86$, the lowest value we ran.

TABLE I. Status of each quantitative result. "Exact" means a consequence of the rules and of the algebra of the deterministic map they define, with no approximation; "mean-field" means a fluctuating belief has been replaced by its mean, or a graph treated as locally tree-like, so that the statement is a prediction about the dynamics whose accuracy must be measured; "numerical" means measured, with N the number of independent realisations.

result	status	qualification
Participation $p_j = x_{I_j}^{a_j}$, Eq. (1); policy $x_{I_j}^*$, Eq. (2)	exact	consequences of rules (i)–(v); Eq. (2) is myopically optimal given $\hat{p}$, not optimal over trajectories
Pair saddle-node p^*, R_c, Eq. (4); validity $a \leq 1/c$, Eq. (5)	exact	tangency of the symmetric deterministic return map
Attention division $1/(k+a)$, Eq. (7)	exact	for priorities drawn afresh per offer, which is the network rule; the correction when they are not is Eq. (14)
Critical degree k_c, Eq. (8); group degree d_c, Eqs. (12) and (13)	exact tangency, mean-field dynamics	the tangency is exact for the mean map; that a coupled state *exists* for $k < k_c$ does not imply the dynamics stays in it. For $m \geq 3$ it does not, and on a network at $\epsilon = 0$ it does not at any cost
Percolation threshold, Eqs. (9) and (10)	mean-field	configuration-model and locally tree-like; overestimates the stationary threshold by $\simeq 1.9$ (cavity) and $\simeq 1.3$ (Molloy–Reed). Locating that threshold needs runs of 6.4×10^6 steps; shorter runs understate it (Sec. VI A)
Refresh correction q, Eq. (14)	approximate	stationary flux balance for a singly-committed agent
Occupancy ϕ, tail indices α, bimodality, recovery fractions, $k_{\min}$, H^*	numerical	pair; N given in each caption
Coupled-edge fraction, S, degree-resolved survival, collapse times	numerical	network and hypergraph; N given in each caption
Membership of the directed-percolation class	untested analogy	used only to predict the *sign* of the mean-field error; no exponents measured, no scaling collapse attempted

At $L = 2$ the level-1 boundary measured the same way lies between $R_I = 0.86$ (0.00 coupled) and $R_I = 0.88$ (0.97), above the saddle-node value $R_c = 0.8332$; the offset is the same metastability quantified in Sec. V F, since at $\tau_{\text{mem}} = 50$ the mean collapse time is only $4.7{\times}10^3$ steps.

The effect is markedly asymmetric in a way that inverts the $\max_j$ rule of the non-adaptive model: at $L = 3$ and $R_I \in \{0.94, 0.96, 0.98\}$, pairs of level-1 agents collapse while a level-2/level-1 pair remains coupled (Fig. 3). The outcome is deterministic at this operating point: all 40 realisations agree in every bar of Fig. 3, and the only quantity that distinguishes promoting one agent from promoting both is the participation propensity $\bar{p}$. In the non-adaptive model the more distracted agent sets the exponent; in the adaptive model the more sophisticated agent sets the phase.

C. Persistence across silence

We next decouple the rate of *opportunity* from willingness by allowing an interaction to be possible only with probability ν per step, and by making the partner's action unobservable except when an interaction occurs. The mean gap between opportunities, $1/\nu$, then sets the rate at which information about the partner arrives.

If unobserved steps decay the estimate — silence treated as refusal — the coupled phase is lost whenever the estimate comes to be dominated by accumulated silence rather than by the last observed encounter. At $1/\nu = 100$, pairs with $\tau_{\text{mem}} \in \{5, 50\}$ remain coupled while $\tau_{\text{mem}} \in \{500, 5000\}$ collapse.

The decisive variable, however, is not the ratio of timescales but whether the estimate is held at all. If unobserved steps leave $\hat{p}$ unchanged, coupling survives at *every* opportunity rate and every memory constant tested, with the realised interaction rate tracking ν to three digits (Table II). A partner model maintained through intervals containing no information is what makes rare interaction viable.

D. Loss of universality

Away from R_c the adaptive system is not scale-free: it polarises to $\tau = 1$ in the coupled phase and $\tau \to \infty$ in the solitary one, with apparent tail indices in the range 4–

TABLE II. Occupancy at $L = 3$, $R_I = 0.99$, level 2, with the partner's action unobservable between interactions. Left: silence decays the estimate. Right: the estimate is held. Mean ± standard error over $N = 40$ independent realisations of 2×10^5 steps each; the first submission reported a single realisation per entry.

	decaying		held	
$1/\nu$	$\tau_{\mathrm{mem}} = 50$	5000	$\tau_{\mathrm{mem}} = 50$	5000
10	0.849(1)	0.000(1)	1.000(1)	1.000(1)
100	0.760(1)	0.000(1)	1.000(1)	1.000(1)
333	0.755(1)	0.000(1)	1.000(1)	1.000(1)

40. Heavy tails reappear only in a narrow window above R_c, where the two fixed points nearly coincide and belief noise drives intermittent switching. Inside that window the interevent coefficient of variation reaches 8–50 and the tail is favoured over an exponential by log-likelihood differences of 10^3 to 10^5, using the tail-comparison procedure of Ref. [22].

The window exists only for $\epsilon > 0$. At $\epsilon = 0$ the solitary phase is absorbing, the pair falls once and never returns, and no value of R_I produces a heavy tail.

Crucially, the exponent measured in the window is *not* given by $\alpha = 1 + 1/\max_j(L_j - 1)$. Because the transition is sharp, ϕ cannot be tuned through R_I; varying only the noise realisation at a fixed operating point, however, spreads ϕ over a wide range, and α tracks it (Table III).

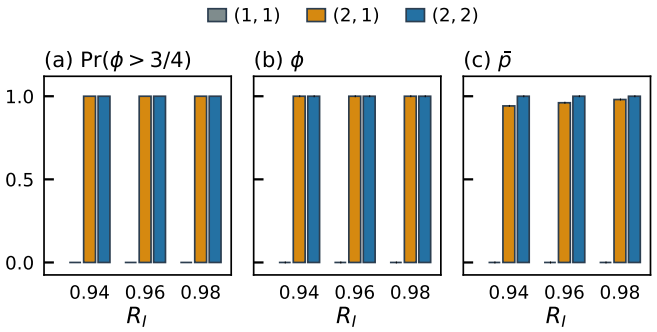


FIG. 3. Asymmetric pairing at $L = 3$, $\tau_{\mathrm{mem}} = 50$, $\epsilon = 0$, 2×10^5 steps, $N = 40$ independent realisations per bar. Bars are grouped by the value R_I of the interaction and, within a group, by the modelling levels (A, B) of the two agents. A single level-2 agent holds the pair in the coupled phase. (a) Fraction of realisations ending coupled, $\phi > 3/4$; (b) occupancy ϕ; (c) time-averaged participation propensity $\bar{p}$. The axis runs a little below zero so that a bar of height zero still shows its outline: the level-1 pairs $(1, 1)$ collapse in every realisation, at every R_I. The outcome is otherwise deterministic, all 40 realisations agreeing, so the error bars are invisible except on $\bar{p}$ — and $\bar{p}$ is the only thing promoting the second agent changes, which is the content of the $(2, 1)$ against $(2, 2)$ comparison. The column headed "occupancy" in the corresponding table of the first submission was in fact $\bar{p}$; the two are shown separately here.

The correlation between ϕ and α is +0.67, with a 95% bootstrap interval of +0.56 to +0.78 over 45 realisations, at $L = 2$, and +0.79 (+0.70, +0.88) over 50 at $L = 6$; the two $\alpha(\phi)$ curves nearly coincide, agreeing to 0.25 and 0.23 in the two ranges of ϕ for which both are measured, while the non-adaptive prediction would require 2.00 and 1.20 respectively, a separation of 0.80.

We conclude that α is not an order parameter of the adaptive system. The occupancy ϕ is the natural replacement: bounded, directly measured, and monotone in R_I, L and modelling level, with the scale-free interevent statistics appearing as a symptom of intermediate ϕ rather than as its definition.

E. Bimodality and irreversibility

At fixed parameters ($L = 2$, $c = 0.35$, $R_I = 0.8428$, $\epsilon = 10^{-2}$, $\tau_{\mathrm{mem}} = 200$, $\hat{p}(0) = p^*$, 3×10^5 steps, occupancy on the final 20%) the outcome over 1000 realisations is strongly bimodal: 264 end with $\phi < 0.25$, 718 with $\phi > 0.75$, and only 18 in between. The Wilson intervals on the two modes are 0.238–0.292 and 0.689–0.745; the 140 realisations of the first submission gave 45, 93 and 2, which these contain.

The outcome is nevertheless not fixed by early noise. Classifying the final phase from a trajectory prefix barely exceeds the majority-class baseline of 0.726: accuracies are 0.726, 0.746, 0.758, 0.776, 0.804 and 0.870 for prefixes of 1, 3, 10, 30, 60 and 120×10^3 steps. Those are in-sample optima, with the threshold chosen on the same data it is scored on. Fitting the threshold on a random half of the realisations and scoring it on the other half, averaged over 400 splits, gives 0.726±0.014, 0.743±0.015, 0.757±0.013, 0.772 ± 0.013, 0.801 ± 0.013 and 0.867 ± 0.011, so the in-sample figures are not appreciably optimistic. The first 10^4 steps carry essentially no information. Collapse is a rare event available throughout the trajectory rather than a bifurcation resolved at the outset.

Collapse is, however, irreversible without intervention: none of the 264 collapsed realisations recovered when left alone, a 95% upper bound of 1.4% on the spontaneous recovery probability over the second half of a 3×10^5-step run. Promoting agents to level 2 at the midpoint recovers 33.7% (89 of 264, 95% confidence interval 28.3–39.6%) of them, whereas pairs of level-2 agents run from $t = 0$ on the same seeds never collapse at all (Table VI). Promoting both agents is worth nothing over promoting one: recovery is driven by a single agent participating against the advice of its own estimate, and the partner need only be able to respond.

F. Collapse as a rare event: survival and first-passage times

That collapse is available throughout the trajectory can be made quantitative, and doing so also settles

TABLE III. Tail index versus occupancy at fixed operating points, varying only the random seed. Rows at $\phi = 1$ have CV $\lesssim 0.8$ and are not heavy-tailed; their fitted indices are reported for completeness only. Sixty realisations of 1.2×10^6 steps were run at each operating point, of which 45 and 50 produced the ≥ 2000 interevent times the tail fit requires; the rest are in the solitary phase and are discarded. The interval on the correlation is a 95% bootstrap interval over realisations, and the last two rows are means over the realisations in that range of ϕ, with their standard errors.

	$L = 2$, $c = 0.35$	$L = 6$, $c = 0.10$
$1 + 1/\max_j(L_j - 1)$	2.00	1.20
realisations N	45	50
$\mathrm{corr}(\phi, \alpha)$	+0.67	+0.79
95% interval	(+0.56, +0.78)	(+0.70, +0.88)
α at $\phi \simeq 0.22$	1.79	—
α for $0.35 \leq \phi < 0.8$	4.51(2)	4.26(2)
α for $\phi \geq 0.8$	4.60(1)	4.37(1)

how much of what we report depends on how long we ran. We declare a pair collapsed at the first step from which its participation propensity stays below $p^*/2$ for a whole memory time — the occupancy criterion turned into a sustained first-passage condition, so that single-step excursions do not count — and record that time T_c. Most realisations have not collapsed when a run ends, so the sample is right-censored and the mean is estimated by the censored exponential maximum likelihood, $\langle T_c \rangle$ = (total exposure)/(number of collapses), whose relative standard error is $1/\sqrt{k}$ with k the number of collapses.

At the reference point ($L = 2$, $c = 0.35$, $R_I = 0.8428$, $\epsilon = 10^{-2}$, $\tau_{\mathrm{mem}} = 200$), 114 of 500 realisations collapse within 2×10^5 steps and $\langle T_c \rangle = (7.8 \pm 0.7) \times 10^5$ steps. The escape is memoryless, by three independent tests. The survival curve is exponential [Fig. 4(a)]. The observed first-passage times are spread almost uniformly across the run, with a coefficient of variation of 0.54 against the $1/\sqrt{3} = 0.577$ expected of an exponential law seen through a window short compared with its mean. And a single rate, fitted to all run lengths at once, reproduces the fraction collapsed over a factor of 32 in run length: 0.032, 0.048, 0.100, 0.200, 0.396, 0.620 measured against 0.029, 0.057, 0.111, 0.210, 0.377, 0.611 predicted [Fig. 4(b)]. Collapse is a Poisson event with a constant hazard, not a bifurcation resolved at the outset — which is what the prefix classification above showed indirectly.

Two things follow, one about the model and one about the measurements. The coupled phase near the transition is metastable, with a lifetime that rises very steeply as the belief noise is reduced: $\langle T_c \rangle$ is 9.2×10^2, 4.7×10^3, 3.2×10^4 and 9.0×10^5 steps at $\tau_{\mathrm{mem}} = 25$, 50, 100, 200, and no collapse at all occurs in 250 runs of 2×10^5 steps at $\tau_{\mathrm{mem}} = 400$ or 800, bounding $\langle T_c \rangle > 5 \times 10^7$ [Fig. 4(c)]. Three decades of lifetime for one decade of memory time is the signature of a noise-activated escape, whose rate is exponentially small in the number of observations the estimate averages; with four usable points we do not fit the functional form. It is the same mechanism that displaces the network threshold in Sec. VI. The measurement consequence is that an occupancy taken over a finite run is not a stationary quantity here: the same operating point gives $\phi = 0.988$ at $T = 2.5 \times 10^4$ and $\phi = 0.649$ at $T = 8 \times 10^5$, while $\langle T_c \rangle$ stays at (8–10) $\times 10^5$ throughout (Supplemental Table I). The lifetime is run-length independent and the occupancy is not, so we state the run length with every occupancy we quote and never compare occupancies measured over different lengths.

Supplemental Table I collects the sensitivity of these quantities to the parameters. The initial belief matters exactly as bistability requires and not otherwise: runs started below the unstable fixed point of $\mathcal{F}$, at $p = 0.521$ here, collapse without exception; runs started above it agree to three digits for $\hat{p}(0) = 0.70$, 0.85 and 1.00; and only the marginal value 0.50, just under the boundary, gives a mixture. The exploration rate acts in two opposite directions: at $\epsilon = 0$ the solitary state is strictly absorbing and 77% of realisations end there, while raising ϵ to 0.03 suppresses collapse altogether, since exploration also feeds the estimate that sustains participation. The intermittent window of Sec. V D therefore sits between the two, at small but nonzero ϵ. The remaining parameters c and L are absent because they enter only through R_c and p^*; the test of their effect is whether the measured boundary moves with them as Eq. (4) says, and Sec. V G reports that it does.

G. The phase boundary in c and L

Neither the cost c nor the queue length L enters the pair dynamics except through R_c and p^*, Eq. (4), so the way to test their effect is to ask whether the measured boundary moves with them as that equation says. Table IV scans R_I across R_c for six (L, c), covering a factor of five in c and both queue lengths for which the bifurcation is interior at these costs, with 40 realisations of 2×10^5 steps at each of 9 values of R_I spanning $R_c - 0.02$ to $R_c + 0.08$.

The agreement is sharper than we expected. Below R_c not one realisation of the 40 is coupled, at any of the six

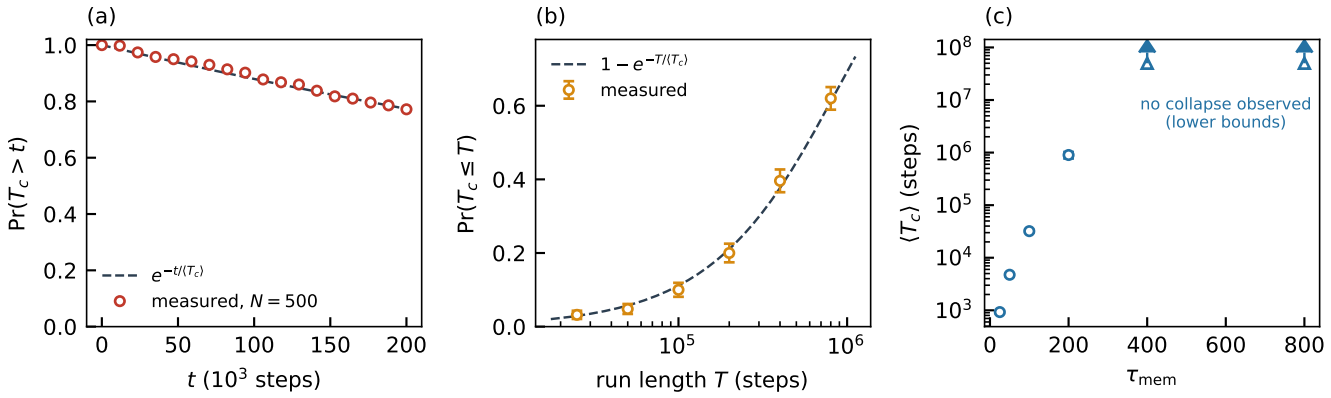


FIG. 4. Escape from the coupled state is a rare event with a constant hazard. Reference point $L = 2$, $c = 0.35$, $R_I = 0.8428$, $\epsilon = 10^{-2}$, $\tau_{\text{mem}} = 200$. Symbols are measurements throughout; dashed lines are the exponential law with $\langle T_c \rangle$ taken from the censored maximum likelihood. (a) Probability that the collapse time exceeds t, over $N = 500$ independent realisations. (b) Probability that it does not exceed the run length, against that run length, $N = 250$ per point, compared with the single rate pooled over all six lengths; error bars are binomial. (c) Mean collapse time against the memory time, $N = 250$ per point; open triangles are lower bounds at $\tau_{\text{mem}} = 400$ and 800, where no realisation collapsed.

TABLE IV. Measured phase boundary against Eq. (4). For each (L, c), R_I is scanned over 9 values spanning $R_c - 0.02$ to $R_c + 0.08$ with $N = 40$ realisations of 2×10^5 steps each, at $\tau_{\text{mem}} = 200$, $\epsilon = 10^{-2}$, $\hat{p}(0) = 0.85$. $R_{1/2}$ is the value of R_I at which half the realisations are still coupled at the end of the run.

L	c	R_c	p^*	N	$R_{1/2}$	$R_{1/2} - R_c$
2	0.20	0.6944	0.4472	40	0.7022	+0.0078
2	0.35	0.8332	0.5916	40	0.8408	+0.0076
2	0.50	0.9142	0.7071	40	0.9200	+0.0058
3	0.10	0.7772	0.3420	40	0.7811	+0.0038
3	0.20	0.9052	0.5429	40	0.9085	+0.0033
3	0.30	0.9651	0.7114	40	0.9678	+0.0026

parameter pairs and at every offset down to $R_c - 0.005$; at R_c itself none is coupled either; and by $R_c + 0.010$ the coupled fraction is between 0.82 and 1.00. Because the coupled state is metastable the boundary still has to be defined operationally, and we take the R_I at which half the realisations are still coupled at the end of the run. It exceeds R_c by 0.0026 to 0.0078 while R_c itself moves from 0.694 to 0.965 across the six rows, so Eq. (4) is confirmed to better than 1% over a range of 0.27 in R_c. The small positive offset is the finite-time one of Sec. V F: just above R_c the two fixed points have barely separated and the lifetime is short, so a run of 2×10^5 steps cannot see a coupled state that does not survive that long. For the sensitivity question the conclusion is that c and L act on the pair only through Eq. (4), which is why Supplemental Table I leaves them out.

H. Patience, not reachability

The partial recovery in Table VI might indicate states from which no policy in this family can return. It does not. As specified above, the level-2 rollout evaluates a single step of unreciprocated participation. We therefore introduce a commitment length k: the number of consecutive steps of unilateral participation the agent evaluates, and latches onto if it decides to proceed.

The decision is deterministic given beliefs, so the reachable region can be mapped directly. Table V reports the smallest $\hat{p}$ at which an agent still chooses to participate, taking $\hat{s} = \hat{p}$. At $k = 1$ the agent participates only if it already believes its partner participates $\gtrsim 46\%$ of the time — a condition a collapsed pair fails by construction. Increasing k alone is insufficient, and increasing the lookahead H alone changes nothing at all: recovery is flat at 35.7% for $H \in \{10, \ldots, 1000\}$ at $k = 1$, at both $\gamma = 0.97$ and $\gamma = 0.999$. Only when a long commitment is combined with a discount horizon $1/(1-\gamma)$ long enough to see past it does the threshold collapse to zero, and then every collapsed realisation recovers.

The residual 35.7% is not policy-driven: it is the fraction of realisations in which exploration lifts $\hat{p}$ above the participation threshold and the agent latches. No state is beyond recovery; the cap measures the reluctance of a short-horizon agent to persist without evidence. The collapsed subsets behind Tables VI and V are different, coming from runs of 3×10^5 and 2×10^5 steps, so the two are not the same measurement; they do agree, 33.7% (28.3–39.6) against 35.7% (25.5–47.4) for the same intervention.

TABLE V. Smallest belief $\hat{p}$ at which the agent still participates (left), and measured recovery of the collapsed realisations (right), versus commitment length k and discount γ. The left half is a property of the rollout and is deterministic; the right half is measured over the $N = 70$ realisations, out of 400, that collapsed as level-1 pairs, with a 95% Wilson interval of 25.5–47.4% on every cell but the last, which has 94.8–100%. Recovery is 35.7% in every cell but one.

	$\min \hat{p}$		recovered	
k	$\gamma = 0.97$	$\gamma = 0.999$	$\gamma = 0.97$	$\gamma = 0.999$
1	0.463	0.458	35.7%	35.7%
10	0.458	0.453	35.7%	35.7%
100	0.428	0.313	35.7%	35.7%
1000	0.428	10^{-6}	35.7%	100%

I. Endogenous commitment

The commitment length above is imposed. Letting the agent evaluate its rollout at each candidate length and latch onto the argmax makes it a property of its beliefs. The argmax is uninformative, pinning to the largest candidate since further length is free once the partner participates, so we characterise the policy by the minimum patience that beats not persisting,

$$k_{\min}(\hat{p}) = \min\{\, k : \ \mathcal{R}(k) > \mathcal{R}(0) \,\}, \tag{6}$$

with $\mathcal{R}(k)$ the rollout value under a k-step commitment.

Two regularities emerge, both in units of the memory constant τ_{mem}. First, $k_{\min}$ does not diverge as $\hat{p} \to 0$; it saturates, and bisecting at $\hat{p} = 10^{-6}$ gives $k_{\min}/\tau_{\text{mem}} = 0.740, 0.740, 0.740, 0.738$ for $\tau_{\text{mem}} = 50, 100, 200, 400$. The price of recovery is therefore a fixed multiple of a memory time and is independent of how unfavourable the current estimate is. The partner begins to respond earlier, at $0.347\,\tau_{\text{mem}}$, where its myopic priority turns positive at $s > c/(R_I + c) = 0.293$; persistence must continue about twice that long before the gain covers the commitment.

Second, whether the agent will pay that price is set by its horizon in the same units: the smallest $H^* = 1/(1-\gamma)$ at which an agent with $\hat{p} = 10^{-6}$ still commits is $H^* \simeq 1.5\,\tau_{\text{mem}}$ (measured 1.50 at $\tau_{\text{mem}} = 100, 200, 400$; the value 2.00 at $\tau_{\text{mem}} = 25, 50$ is an upper bound set by the resolution of the H ladder).

These predict the simulation. At $\tau_{\text{mem}} = 200$, so $H^* \simeq 300$, an agent choosing its own commitment length recovers 35.7% of collapsed realisations at $\gamma \leq 0.995$ ($H \leq 200$) and 100% at $\gamma = 0.999$ ($H = 1000$), while the fixed $k = 1$ control stays at 35.7% throughout. These are the same 70 collapsed realisations as Table V, so the two experiments are directly comparable; the 95% Wilson intervals are 25.5–47.4% and 94.8–100%. Judging recovery by interaction rate instead — unbiased, since ϕ exceeds 1/2 whenever one agent is latched whatever its partner does — moves the baseline to 70.0% (58.5–79.5) but leaves the contrast unchanged, with the mean rate rising from 0.277 to 0.616 and the partner's payoff from 0.610 to 0.725 against a solitary $\langle x_O \rangle = 0.5$. The recoveries are genuine.

TABLE VI. Intervention on the 264 realisations that collapsed as level-1 pairs, with promotion at the midpoint of the run. The interval is the 95% Wilson interval on the recovered fraction; the mean final occupancy is given with its standard error.

intervention	recovered	mean final ϕ
none (control)	0.0% (0.0–1.4)	0.026(2)
agent $A \to$ level 2	33.7% (28.3–39.6)	0.350(29)
both $\to$ level 2	33.7% (28.3–39.6)	0.350(29)
level 2 from $t = 0$	100% (98.6–100)	1.000

The agent that restores the interaction is not the one that gains most from it: the committing agent earns 0.665 against its partner's 0.725, the difference being the unreciprocated offers it absorbs (before the recovery the two earn 0.589 and 0.610, so the intervention costs the committing agent the smaller share of what it creates).

VI. PERCOLATION ON NETWORKS

The two-agent model extends to a network by giving agent i one interacting task per neighbour alongside its solitary aggregate, with a single item selected per step. Attention then becomes a scarce resource: an agent with k relations leaves $k-1$ offers unanswered at every step, and each relation must compete with the others as well as with the private queue. An edge executes, as before, only when both endpoints select it.

Two amendments to rule (ii) are needed, since abandoning either produces an artefact. First, the interacting-task priority must be *drawn* from a density the policy controls, $u\,x_{I_j}$ with u uniform, rather than being a deterministic function of $\hat{p}$; otherwise each agent locks onto its single best partner, every other relation starves, and the coupled subgraph is forced to be a perfect matching. Second, an *offer* must consume the attempt whether or not it is reciprocated. Redrawing only on execution freezes the choice: i offers to the same partner indefinitely, and if that partner's best is not i the pair deadlocks.

The topology is fixed here and only the state of each edge evolves, but the coupled subgraph is itself an effective network that grows and fragments, so the setting is adjacent to the literature on adaptive networks [23], where the interplay of node dynamics with topology generates discontinuous transitions and bistability of the kind seen below.

A capacity limit on the number of interactions a node may sustain has been imposed before in a game-theoretic setting: restricting the associative capacity of nodes in a prisoner's dilemma on scale-free networks changes the

level of cooperation and admits an optimal capacity [24]. The distinction here is that no capacity is imposed. It follows from Eq. (7), because an agent executing one task per step divides its attention as $1/(k+a)$, and the resulting k_c is a property of the queue rather than a parameter. The game is also different — coordination with an absorbing failure state rather than a dilemma — so the two capacities are not the same quantity, but they act on the same variable.

We use three growth rules. Triadic closure $LS(\ell = 1)$ is the undirected local-search rule in which a newcomer links to a random node and to one of its neighbours, closing a triangle at every step in the manner long argued to organise social ties [25]; growth by triadic closure generates community structure spontaneously, and randomising the attachment destroys it [26]. Against it we set the degree-preserving randomization of LS, and preferential attachment $BA(m = 2)$ [27] as a non-local control. Randomizing LS preserves the degree sequence exactly, so any difference between the two is attributable to local structure alone.

We use c, the cost of an unreciprocated offer, as the control parameter, and call an edge coupled when its interaction rate exceeds ten times the solitary baseline ϵ^2. The order parameter S is the largest connected component of the coupled subgraph as a fraction of n (Table VII).

The solitary phase percolates between $c = 0.025$ and $c = 0.030$, and the interesting behaviour is at the lower edge. There the two networks have statistically identical coupled edge fractions, 0.517(2) against 0.522(2) over 24 realisations each, and degree-resolved survival that agrees to within 0.011 in every degree bin (Table VIII), yet entirely different connectivity: the triadic-closure network has $S = 0.227 \pm 0.020$ and 36.3 components of three nodes or more, while its randomization retains $S = 0.831 \pm 0.005$ in 9.8 components. The same contrast appears one step earlier, at $c = 0.020$, in weaker form — $S = 0.882 \pm 0.022$ against 1.000 at coupled edge fractions 0.840 and 0.849 — so it is not an artefact of one value of c. Local structure does not determine which relations survive; it determines how their failure is arranged. Under triadic closure the coupled phase fragments into separated clusters, whereas at identical degrees without local structure it thins while staying connected. The split between local and non-local rules that governs the emergence of communities [26] thus reappears as a split in the way the solitary phase invades.

Survival falls monotonically with degree — the fraction of a node's relations that remain coupled at $c = 0.020$ is 0.972, 0.889, 0.726 and 0.457 for degrees near 2, 4, 8 and 16 — and the two networks agree to within 0.03 at every degree and to within 0.01 for $k \lesssim 8$; at $c = 0.025$, where their connectivities differ most, they agree to within 0.011 at every degree (Table VIII). This follows directly from the attention constraint: $\hat{p} \simeq P(\text{active})/k$, so a relation is sustainable only while $k \lesssim P(\text{active})(R_I + c)/c$. High degree is a liability rather than a protection, because what an agent competes for is its partner's attention, and the coincidence of the two curves shows the effect to be independent of topology.

A. Realisations, error bars, finite size and robustness

The numbers above come from 24 independent realisations per point — a fresh graph as well as fresh dynamics, since the growth rule draws from the same seeded stream — with runs of 4×10^5 steps, ten times longer than in the first submission. What follows is what that buys: a test of the edge classification, the behaviour with system size, and the dependence on run length and initial condition. None of it changes the mechanism; it changes how much of the mechanism we are entitled to claim.

The coupled-edge criterion. Calling an edge coupled above a rate of $10\epsilon^2$ is not a cut through a smooth distribution. The per-edge rates are bimodal [Fig. 5(b)]: a solitary edge executes only when both endpoints explore, which places it between 10^{-5} and 2×10^{-4}, while a coupled edge executes between 10^{-3} and 3×10^{-2}. The region between is empty deep in either phase and holds only 14% of edges at $c = 0.025$, in the middle of the transition. Moving the cut across that gap leaves the order parameter alone: Supplemental Table III varies it over the decade from 3×10^{-4} to 3×10^{-3}, with $10\epsilon^2$ at the centre, and S moves by less than 0.012 in the coupled phase and stays below 0.011 in the solitary one. Outside that decade it fails, and it is worth saying how. At a cut of ϵ^2 the exploration floor percolates by itself, giving a spurious $S = 0.499 \pm 0.013$ at $c = 0.070$ where no edge is coupled; at 5×10^{-3} the cut starts deleting coupled edges, and S at $c = 0.020$ falls from 1.000 to 0.849(8).

System size. Supplemental Table IV and Fig. 6 repeat the measurement at $n = 300$ to 4800. The randomized network shows no size dependence at all: S is 1.000 for $c \leq 0.020$, 0.82–0.83 at $c = 0.025$ and of order $1/n$ at $c = 0.030$, at every size. The transition therefore sits between $c = 0.025$ and $c = 0.030$ independently of n, and what vanishes above it is a giant component rather than merely a large one: the largest coupled cluster at $c = 0.030$ holds between six and ten nodes at every size.

Under triadic closure the size dependence is strong, and it runs the other way. At $c = 0.020$, S falls from 0.936(24) at $n = 300$ to 0.370(77) at $n = 4800$, and at $c = 0.025$ from 0.305(22) to 0.049(8); over that range $S \sim n^{-0.17}$, $n^{-0.32}$ and $n^{-0.66}$ at $c = 0.015$, 0.020 and 0.025. The coupled edge fraction meanwhile matches the randomization to within 0.005 at every size and every cost, so nothing about *which* relations fail is changing. We read this as the coupled subgraph of the locally structured network ceasing to percolate in the large-system limit at costs where its randomization still percolates: local structure lowers the percolation threshold of the solitary phase rather than rearranging a fixed amount of damage at one size. We put it no more strongly, since

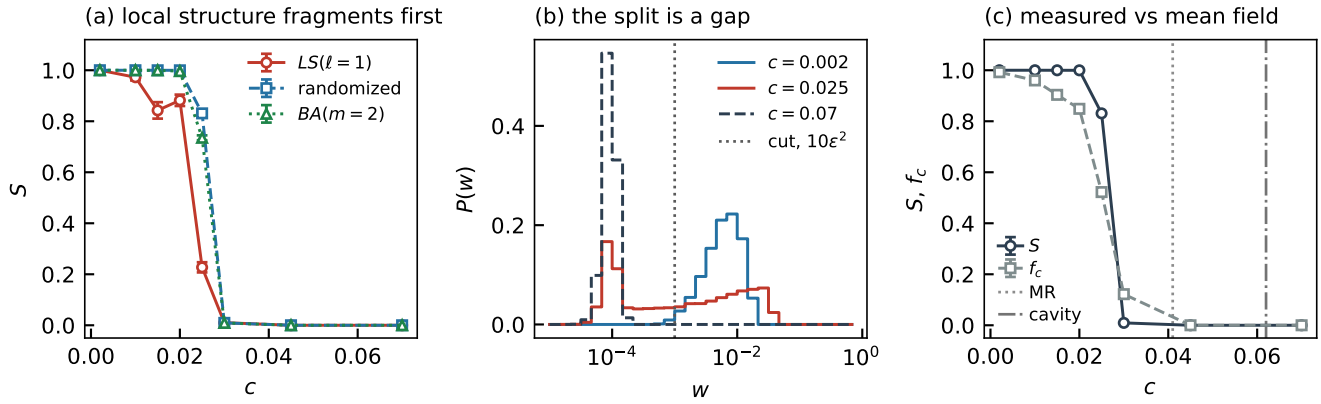


FIG. 5. Percolation of the solitary phase. All panels: $n = 600$, $\langle k \rangle = 4$, $R_I = 0.95$, 4×10^5 steps, $N = 24$ independent realisations per point, error bars the standard error over realisations. (a) Giant component S of the coupled subgraph against the cost c of an unreciprocated offer. At $c = 0.025$ the three rules carry the same coupled edge fraction f_c, to within 0.006, but only the local rule fragments. (b) Distribution $P(w)$ of the per-edge interaction rate w, pooled over realisations of $LS(\ell = 1)$. A collapsed edge executes only when both endpoints explore, which places it near 10^{-4}; a coupled edge executes two orders of magnitude more often. The classification cut $10\epsilon^2$, dotted, lies in the gap between the two bands. (c) Coupled edge fraction f_c and S on the randomized network, with the two mean-field thresholds marked, Molloy–Reed (MR, dotted) and cavity (dash-dotted): the transition lies between $c = 0.025$ and $c = 0.030$, against $c_c = 0.041$ from Eq. (9) and $c_c = 0.062$ from Eq. (10).

TABLE VII. Coupled edge fraction and giant component S of the coupled subgraph, at $R_I = 0.95$, $n = 600$, $\langle k \rangle = 4$, 4×10^5 steps, mean ± standard error over $N = 24$ independent realisations. The first submission reported three realisations of 4×10^4 steps; the transition has moved from between $c = 0.020$ and 0.030 to between $c = 0.025$ and 0.030, and the row at which the growth rules separate has moved with it.

	$LS(\ell = 1)$		randomized		$BA(m = 2)$	
c	edges	S	edges	S	edges	S
0.002	0.990(1)	1.000(1)	0.992(1)	1.000(1)	0.949(3)	1.000(1)
0.010	0.955(2)	0.973(13)	0.960(2)	1.000(1)	0.877(4)	1.000(1)
0.015	0.899(2)	0.843(32)	0.904(2)	1.000(1)	0.804(4)	1.000(1)
0.020	0.840(2)	0.882(22)	0.849(2)	1.000(1)	0.728(4)	0.996(1)
0.025	0.517(2)	0.227(20)	0.522(2)	0.831(5)	0.466(3)	0.735(10)
0.030	0.123(2)	0.010(1)	0.122(2)	0.009(1)	0.109(2)	0.009(1)
0.045	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)
0.070	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)

five sizes are enough to show that the effect grows with n and not enough to fit a critical exponent, which we do not attempt. What the argument needs is the negative statement: the contrast in Table VII is not an artefact of $n = 600$.

Run length and initial condition. The coupled state is metastable here as for a pair, and Supplemental Table II makes the consequence explicit. The coupled edge fraction rises with the length of the run at every c near the transition, and at $c = 0.025$ is still rising after 1.6×10^6 steps; the giant component follows, 0.020(1) after 2.5×10^4 steps, 0.831(5) after 4×10^5 and 0.982(2) after 1.6×10^6. A threshold read off a run of any finite length is therefore a *lower* bound on the stationary one. The drift is not a memory of the initial condition: starting every belief at $\hat{p}(0) = 0.10$ rather than 0.85 changes no entry in the second block of Supplemental Table II beyond its error bar, so at $\epsilon = 10^{-2}$ the network relaxes to the same state from either side, with no hysteresis at this run length. What is slow is the relaxation itself.

Pushing the runs to 6.4×10^6 steps, sixteen times the length used elsewhere, shows where this stops. At $c = 0.025$ the coupled edge fraction climbs to 0.841(4) and S to 0.998(1); at $c = 0.030$ to 0.415(5) and 0.603(29), so what looks solitary after 4×10^5 steps ($S = 0.009(1)$) is a percolating coupled phase that had not yet formed. At $c = 0.035$, by contrast, the coupled fraction is 0.004(1) at 1.6×10^6 steps and 0.003(1) at 6.4×10^6 — flat, and therefore stationary — and at $c = 0.045$ and 0.062 nothing survives. The stationary transition is thus bracketed

TABLE VIII. Degree-resolved survival: the fraction of a node's relations that remain coupled, for nodes of degree k in $[b, 2b)$, at $n = 600$, $\langle k\rangle = 4$, $R_I = 0.95$, 4×10^5 steps, mean $\pm$ standard error over N realisations. The three growth rules agree, so the effect is a property of the attention constraint and not of the topology; what differs between them is how the failures are arranged, not which relations fail.

	N	$b = 2$	4	8	16
$c = 0.020$					
$LS(\ell = 1)$	24	0.972(1)	0.889(2)	0.726(3)	0.457(9)
randomized	24	0.973(1)	0.898(2)	0.737(4)	0.485(6)
$BA(m = 2)$	24	0.891(4)	0.791(3)	0.688(3)	0.464(8)
$c = 0.025$					
$LS(\ell = 1)$	24	0.677(3)	0.531(3)	0.370(4)	0.240(6)
randomized	24	0.688(3)	0.527(4)	0.379(4)	0.241(5)
$BA(m = 2)$	24	0.616(3)	0.503(4)	0.379(4)	0.240(5)

between $c = 0.030$, still rising at the longest run, and $c = 0.035$, where it has stopped, giving $c_c \simeq 0.032$ against the $c_c \simeq 0.027$ that 4×10^5 steps suggest. Part of what the first submission attributed to fluctuations was the length of the runs.

Sensitivity. The lower block of Supplemental Table II varies ϵ, τ_{mem}, R_I and L one at a time about the reference point, and one of the four matters far more than Eq. (8) allows for. The exploration rate does not move the transition; it decides whether there is a coupled phase at all. At $\epsilon = 0$ no edge is coupled at any c in the table, including $c = 0.010$, where $k_c = 22$ leaves a coupled state comfortably available at $\langle k\rangle = 4$; at $\epsilon = 3 \times 10^{-3}$ the coupled phase survives only to $c = 0.015$; at $\epsilon = 3 \times 10^{-2}$ and 10^{-1} nine tenths of the edges are coupled at every c up to 0.030. Equation (8) treats exploration as a small tax on the attention budget, replacing $c(k + a)$ by $c(k+a)(1-\epsilon)^{-k}$ and moving the predicted threshold from 0.064 to 0.062. The measurement says it does something else: it is what stops a belief from spiralling downward, which a treatment that replaces the belief by its mean cannot see. That, rather than a general appeal to fluctuations, is why the mean-field threshold is wrong here, and it is the network counterpart of the intermittent window existing only at $\epsilon > 0$.

The other three behave as the theory requires. Raising R_I raises the coupled fraction at every c; raising L lowers it sharply, k_c at $c = 0.020$ falling from 10 at $L = 2$ to 4 at $L = 3$ and 1 at $L = 4$ while the measured fraction at $c = 0.010$ falls from 0.960(2) to 0.385(3) to 0.000. The memory time acts mainly through the relaxation rate, a longer memory meaning a slower approach to the stationary state at fixed run length: the coupled fraction at $c = 0.025$ falls from 0.966(4) at $\tau_{\mathrm{mem}} = 50$ to 0.267(1) at 400, then rises to 0.450(2) at 800. That is not monotone, and we do not interpret it beyond noting that at a fixed number of steps memory time and run length are not independent.

Relaxation and the network collapse time. The edge-level analogue of the pair collapse time of Sec. V F is not well defined, for a reason worth recording: the interevent times of a coupled edge are heavy-tailed, so a silence of any fixed length occurs in the coupled phase too, and a death criterion built from one measures the criterion rather than the dynamics. The global execution rate does not have that problem. Below the transition it is stationary — the executed-edge fraction in the last window of a 4×10^5-step run differs from the first by between -3% and $+13\%$ at every $c \le 0.025$, in all three growth rules — and above it the decay has a measurable half-life: $(8.1 \pm 0.1) \times 10^4$ steps at $c = 0.030$, $(5.9 \pm 0.2) \times 10^3$ at $c = 0.045$ and 2×10^3 at $c = 0.070$, ending at the exploration floor in each case. That is the network counterpart of Fig. 4(c).

B. Analytic percolation threshold

The transition admits a closed-form treatment. Since priorities are uX with $u \sim U(0, 1)$ drawn per offer and the solitary task has $F_O(t) = t^a$, the probability that an agent of degree k selects one particular neighbour is

$$\int_0^1 u^{k-1}(uX)^a \, du = \frac{X^a}{k + a}, \tag{7}$$

so attention divides as $1/(k + a)$. Self-consistency $X = \Psi(X^a/(k + a))$ with $\Psi(p) = B - c/p$ and $B = R_I + c$ gives $X^{a+1} - BX^a + c(k + a) = 0$, whose tangency at $X^* = aB/(a + 1)$ fixes a critical degree

$$k_c = \frac{a^a (R_I + c)^{a+1}}{c\,(a + 1)^{a+1}} - a, \tag{8}$$

equal to $(R_I+c)^2/4c-1$ for $a = 1$. Beyond k_c no coupled state exists at any belief: attention is divided k ways, $\hat{p} \sim 1/k$, and it falls below the collapse threshold $c/(R_I + c)$.

An edge is live only if sustained from both ends, so the process is degree-targeted site percolation [28] and the Molloy–Reed criterion applies with the second moment truncated at k_c,

$$\sum_{k \le k_c} k(k - 1)\, p_k > \langle k\rangle. \tag{9}$$

Equation (9) uses the full degree, but a node whose relations have already failed divides its attention only among the survivors. Writing ϕ for the probability that an edge is sustained from one end and $q_k = kp_k/\langle k\rangle$, the cavity equation closes on a locally tree-like graph as

$$\phi = \sum_k q_k \, \mathbb{P}[\mathrm{Bin}(k - 1, \phi) \le k_c - 1] \,, \tag{10}$$

with a giant component iff $b = \sum_k q_k \, \mathbb{E}[(m - 1)\mathbf{1}\{m \le k_c\}]/\phi > 1$, $m = 1 + \mathrm{Bin}(k - 1, \phi)$. Setting $\phi = 1$ recovers Eq. (9). That limit ignores that a node whose

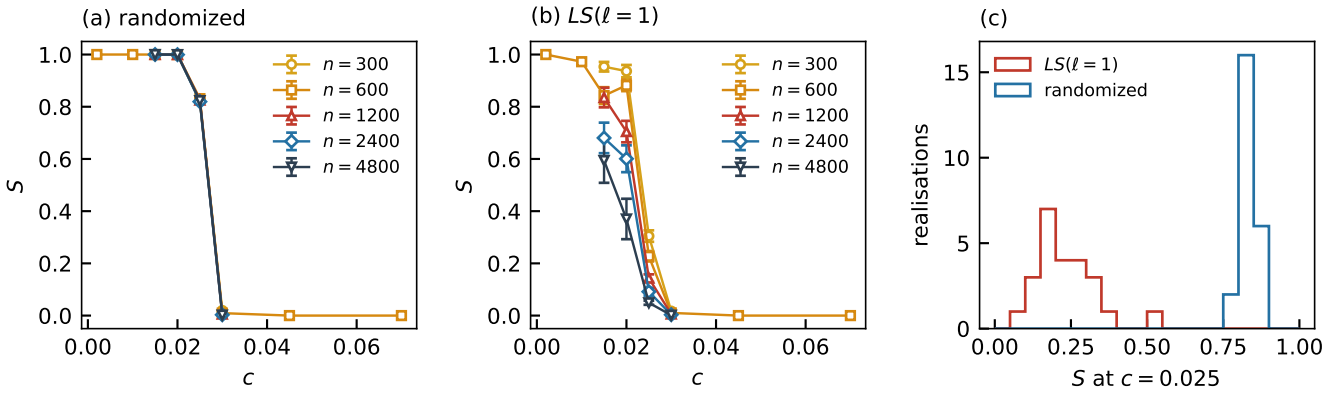


FIG. 6. The network transition with system size. (a) Giant component of the coupled subgraph against c on the randomized network at five sizes; (b) the same for triadic closure, $LS(\ell = 1)$; (c) distribution of S over the $N = 24$ realisations at $c = 0.025$, $n = 600$, where the two rules differ most. Error bars in (a) and (b) are standard errors over realisations, N as in Supplemental Table IV.

relations have already failed divides its attention among the survivors only, so it underestimates how many relations remain sustainable and yields the more conservative threshold: for the degree distributions used here Eq. (9) gives $c_c = 0.041$ against $c_c = 0.062$ from Eq. (10). Both lie above the measured threshold, so both overestimate the extent of the coupled phase; Eq. (10) is the looser of the two and is the one quoted below. (The first submission stated the relation between the two the other way round; the numbers above are the correction.) Exploration at rate ϵ sets $x = 1$ and hence wins the selection, consuming the attention slot with probability $1-(1-\epsilon)^k$; including it replaces $c(k+a)$ by $c(k+a)(1-\epsilon)^{-k}$ in (8).

Equation (8) reproduces the measured degree dependence: survival falls monotonically with k, and $k_c = 10.8$ at $c = 0.020$ sits where the measured survival halves (0.726(3) at $k \simeq 8$ and 0.457(9) at $k \simeq 16$). The threshold itself, however, is a mean-field upper bound. On the randomized network, where the tree-like assumption holds, Eq. (10) gives $c_c \simeq 0.062$ against a measured $c_c \simeq 0.032$. That measurement takes some care, because the coupled phase forms slowly: after 4×10^5 steps the transition appears to lie between $c = 0.025$ and $c = 0.030$, but after 6.4×10^6 steps $c = 0.030$ is percolating and still growing while $c = 0.035$ has stopped changing, which brackets the stationary threshold between the two (Sec. VI A). Equation (9) gives $c_c = 0.041$, so the two mean-field estimates overestimate by factors of 1.9 and 1.3. The discrepancy is the expected one: a mean-field treatment cannot capture fluctuation-driven escape into an absorbing state, and that escape is precisely the stag-hunt mechanism, the runaway in which a falling $\hat{p}$ lowers X, which reduces the offers received, which lowers $\hat{p}$ again. Relations that (8) declares viable are destroyed by it.

The measurements of Sec. VI A make this specific, replacing an appeal to fluctuations by a statement about two rates. Equation (8) asks whether a coupled fixed point exists; the dynamics answers a different question, whether a relation in that state stays there. The escape rate is now measured directly for a pair (Sec. V F) — finite, exponentially small in τ_{mem}, and independent of how long one watches — and the return rate is set by exploration, since two agents must explore in the same step. What is stationary is the balance of the two, and c enters by setting the first.

Removing the second makes the point unambiguous. At $\epsilon = 0$ there is no return, and the coupled phase decays away at *every* cost we tested, down to $c = 5\times10^{-4}$, where Eq. (8) gives $k_c = 450$ and a degree-four node is nowhere near the mean-field boundary: the executed-edge fraction falls from 4.0×10^{-3} to 0.7×10^{-3} within 8×10^3 steps and never recovers, while at $\epsilon = 10^{-3}$ it settles at 3.9×10^{-3} and stays. The threshold in c is therefore not where a coupled state ceases to exist, but where the lifetime of a relation falls below the time exploration needs to rebuild it — which no treatment that replaces a belief by its mean can locate. Equation (8) remains exact as a statement about existence, and remains the right predictor of the *degree* dependence, a statement about attention rather than noise.

Two points follow. Convergence near the transition is slow and directional: the coupled fraction rises with the length of the run at every c near the transition, and the drift is not a memory of the initial condition, both as shown in Sec. VI A. A threshold read off a run of given length is therefore a *lower* bound on the stationary one, and a weak one near the transition: the giant component at $c = 0.025$ is 0.020(1) after 2.5×10^4 steps and 0.982(2) after 1.6×10^6. And the truncation in (9) acts at the *top* of the degree distribution. For $p_k \sim k^{-\gamma}$ with $2 < \gamma < 3$ the second moment diverges and the network is robust to random removal, but the removal here is targeted at high degree — the one attack to which such networks are most fragile. A scale-free contact structure is therefore maximally vulnerable to this form of collapse, for a mechanical rather than a topological reason: what an agent

competes for is its partner's attention, so high degree is a liability. The mathematics is that of intentional attack [16], but the attack is self-generated: no adversary removes hubs, and the familiar contrast between robustness to failure and fragility to targeted attack acquires an endogenous form.

The overestimate is also the expected behaviour of the approximation rather than a defect peculiar to this model. Directed percolation enters here as an analogy and nothing more. The model has the two ingredients that define the class — a single absorbing state and a fluctuating active phase — and for such models a mean-field treatment is known to overestimate the ordered phase below the upper critical dimension four [29], so the direction of our discrepancy is the expected one. That is the entire content of the appeal. Nothing here tests membership: we measure no critical exponents and attempt no finite-size scaling, and the substrate is a network rather than a lattice of definite dimension, so it is not even clear that the below-threshold values are the relevant ones. Establishing a universality class would need an order-parameter scaling collapse and at least two independently determined exponents; we do neither, and every statement we make concerns the location and discontinuous character of the transition, not its critical behaviour.

We report the transition location but claim no exponent. The campaign behind Sec. VI A — five system sizes from $n = 300$ to $n = 4800$, 8 to 24 realisations each, run lengths to 1.6×10^6 steps — is enough to bracket the transition and to show that the bracket does not move with system size, and it is not enough to extract a critical exponent or to collapse an order parameter, which we do not attempt. The agents here are also all level 1, the network analogue of the self-model being left for future work.

VII. GROUPS AND HYPERGRAPHS

Many joint activities are not pairwise. A meeting takes place only if everybody turns up, which makes the interacting task a hyperedge rather than an edge. Structures encoding interactions among three or more units, and the dynamics they support, are reviewed in Ref. [30]. Their behavioural side — how interactions involving more than two people shape collective human outcomes, and how much more fragile coordination becomes when everybody has to be present at once — is reviewed in Ref. [33], and the group limit we obtain below is a queueing-theoretic instance of exactly that fragility. Encoding group interactions on hypergraphs is by now a standard device for evolutionary games in particular, where it changes the replicator dynamics and supplies a mechanism for the survival of cooperation [31, 32]; what follows applies it to a coordination requirement inside a queue. We therefore let agent i hold one interacting task per hyperedge it belongs to, write d_i for that number, and require a hyperedge e of size m to execute only when all m of its members select it in the same step. Everything else is as before.

Selection is unchanged, so Eq. (7) still gives $X^a/(d+a)$, but an agent now needs the other $m-1$ members rather than one, so $\hat{p} = [X^a/(d+a)]^{m-1}$ and self-consistency yields

$$X^{n+1} - BX^n + c\,(d+a)^{m-1} = 0, \qquad n = a(m-1), \quad (11)$$

with $B = R_I + c$. This has the same form as the pairwise case with a replaced by n, so the tangency argument carries over and gives a critical group degree

$$d_c(m) = \left[\frac{n^n B^{\,n+1}}{c\,(n+1)^{n+1}}\right]^{1/(m-1)} - a, \qquad (12)$$

which reduces to Eq. (8) at $m = 2$. The exponent $1/(m-1)$ carries the content: the coordination requirement enters multiplicatively in the number of people who must coincide, so d_c falls very steeply with group size. At $R_I = 0.95$ it drops from 10.76 to 1.60 to 0.67 for $m = 2, 3, 4$ at $c = 0.020$, and from 112 to 6.99 to 2.51 at $c = 0.002$.

Because d_c can fall below one, Eq. (12) implies a *maximum group size* even for an agent that belongs to nothing else. Requiring $d_c(m) \geq 1$ gives $m_{\max} = 2, 3, 4, 5$ at $c = 0.050, 0.020, 0.010, 0.002$. It is worth noting that freely forming conversational groups are observed to cap at about four participants and to fission beyond five [34]. Two capacity arguments have been offered for that limit, one acoustic and attentional [34] and one based on a bound on the number of other minds a participant can model at once [35]. The present mechanism is a third of the same character — attention as a scarce resource in a queue — and we make no claim that it is the operative one.

Simulation confirms Eq. (12) at $m = 2$: the fraction of groups that remain coupled falls from 1.000 at $d = 1$ to 0.309 at $d = 16$, straddling $d_c = 10.76$. Over $N = 12$ independent hypergraph realisations the sequence is 1.000, 0.969(3), 0.765(5), 0.538(4), 0.310(2) at $d = 1, 2, 4, 8, 16$ (Table IX), the parenthesis giving the standard error in the last digit. At $m = 3$ it fails. The coupled fraction is 0.031(5) at $d = 1$ and falls to 0.015(1) at $d = 16$; at $m = 4$ no group survives in any realisation. Equation (12) gives $d_c = 1.60$ and so permits $d = 1$, and the fixed point is genuinely there — solving (11) at $m = 3$, $d = 1$, $c = 0.020$ gives a stable root $X = 0.862$ against an unstable one at $X = 0.363$, and the simulation begins exactly on it, with a measured meeting rate of 0.14 against the predicted $(X/2)^3 = 0.11$. Within a few thousand steps it nevertheless collapses to zero.

The reason is that a group fails if *any single member's* belief drifts below threshold, whereupon that member withdraws, the remaining members observe no meeting, and their estimates follow. Escape from the coupled state is therefore available through m independent

TABLE IX. Fraction of groups that remain coupled, at $R_I = 0.95$, $c = 0.020$, $\tau_{\rm mem} = 200$, $\epsilon = 10^{-2}$, 4×10^4 steps, mean $\pm$ standard error over $N = 12$ independent d-regular m-uniform hypergraphs of 240 agents. The critical group degree of Eq. (12) is $d_c = 10.76$, 1.60 and 0.67 for $m = 2, 3, 4$. It describes $m = 2$ and fails at $m \geq 3$, where groups that the criterion admits are destroyed by fluctuations.

m	$d = 1$	2	4	8	16
2	1.000(1)	0.969(3)	0.765(5)	0.538(4)	0.310(2)
3	0.031(5)	0.021(2)	0.019(2)	0.016(2)	0.015(1)
4	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)

routes rather than one, while the coordination probability itself falls as the mth power. Both effects push the same way, and neither is visible to a mean-field treatment that asks only whether a fixed point exists. The diagnosis is confirmed by suppressing the noise: lengthening $\tau_{\rm mem}$, which reduces the variance of the belief estimate, restores a nonzero meeting rate ($N = 12$: the mean rate at $m = 3$, $d = 1$ is 0.0000(1) at $\tau_{\rm mem} = 200$, 10^3 and 5×10^3 and 0.0060(1) at 2×10^4 for $c = 0.020$, while at $c = 0.005$, where $d_c = 4.08$ leaves a wider margin, it is already 0.0058(1) at $\tau_{\rm mem} = 10^3$ and rises to 0.0081(1) at 2×10^4).

The rescue is only partial — the recovered rates remain an order of magnitude below the mean-field value — so the conclusion is qualitative rather than quantitative. The factor-of-2 discrepancy found in the pairwise network case becomes, for $m \geq 3$, a failure of kind: Eq. (12) predicts viable groups that the dynamics destroys. Group coordination is fragile not because coordinated states fail to exist, but because the absorbing state is reachable through any one participant.

A. Reward proportional to group size

A joint activity need not be worth the same however many take part: often the value of the occasion grows with the number present. Replacing R_I by $R_I(m) = r\,m$ asks whether a reward growing with the group can offset a coordination requirement that tightens with it.

The interior tangency disappears. Since $rm > 1$ already at $m = 2$ for r of order unity, the priority saturates at the clipping bound $x_I^* = 1$ and Eq. (12) no longer applies: the coupled state is created at the boundary, not in a saddle-node. Viability is the condition that the bound be attained, $c(1-\hat{p})/\hat{p} \leq R_I(m) - 1$ with $\hat{p} = [1/(d+a)]^{m-1}$, giving

$$d_c(m) = \left[\frac{rm - 1 + c}{c}\right]^{1/(m-1)} - a. \tag{13}$$

A numerical solution of the full fixed-point map agrees with Eq. (13) exactly, returning $X^* = 1$ for every m it admits.

This raises the ceiling substantially: at $r = 0.95$ the maximum group size for a singly-committed agent rises from $2, 3, 4, 5$ to $8, 9, 10, 13$ at $c = 0.050, 0.020, 0.010, 0.002$. It does not remove it. Since $(d+a)^{m-1} \leq (R_I(m) - 1 + c)/c$ pits an exponential against a linear function, $d_c \to 0$ as m grows, and arbitrarily large groups would need $R_I(m) \gtrsim c\,(1+a)^{m-1}$ — exponential in group size. No polynomial increase in the value of an occasion compensates for the cost of getting everybody free at once.

The simulations show a smaller gain still, and the discrepancy is instructive. Equation (7) was obtained assuming that the priority of each task is drawn afresh at every step. In the queueing substrate it is not: a priority is redrawn only when its task is acted on, so a membership that is seldom selected is seldom refreshed and its low priority persists. This is the same mechanism that generates heavy-tailed waiting times in the original model [1, 6], and here it depresses participation. Balancing the flux with a redraw rate $u + \epsilon$ gives a stationary density $\pi(u) \propto (u+\epsilon)^{-1}$, so a singly-committed agent participates with probability

$$q = \frac{1 - \epsilon \ln[(1+\epsilon)/\epsilon]}{\ln[(1+\epsilon)/\epsilon]}, \tag{14}$$

equal to 0.207 at $\epsilon = 10^{-2}$ rather than the 1/2 that a freshly drawn priority would give. The measured meeting rate at $m = 2$, $d = 1$ is 0.0463(1) over $N = 12$ realisations against $q^2 = 0.0427$, where the fresh-priority value 1/4 is wrong by a factor of five. Replacing $\hat{p}$ by q^{m-1} in the viability condition moves the ceiling to $m_{\max} = 4$, and the simulations have $m = 2$ robust (0.0463(1)), $m = 3$ marginal at a rate of 10^{-3} (0.0010(1)), and $m \geq 4$ extinct (0.0000(1) at $m = 4, 6, 8, 10$).

The correction matters here because a singly-committed agent has one priority to refresh; an agent holding many selects the largest, which is refreshed often, so Eq. (7) is accurate where it was used above and worst in precisely this case. The direction is unaffected: a reward proportional to group size raises the largest sustainable group but does not abolish the limit, and raises it by less than the existence criterion promises. What bounds the group is not the worth of the occasion but the rate at which everybody is free at once.

VIII. CONCLUSIONS

Letting agents choose the priority of a task that requires someone else turns a stochastic process into a system with a phase transition, and the transition organises everything else. Where the non-adaptive queueing models deliver one number, an interevent exponent from a small family of universality classes, the adaptive model delivers a phase diagram, and the exponent survives only as a symptom of where the system sits on it.

Its nature. The coupled and solitary phases are separated by a saddle-node bifurcation, Eqs. (4), (8) and

(12), obtained in closed form in all three settings. It is discontinuous and the solitary phase is absorbing.

The conditions that sustain joint activity are not the conditions that restore it: no collapsed pair recovered spontaneously, recovery requires unilateral persistence for $0.74\,\tau_{\mathrm{mem}}$, and only an agent valuing the future beyond $1.5\,\tau_{\mathrm{mem}}$ pays it. A relation is cheap to keep and expensive to rebuild, at a price set by how long it takes to be forgotten.

What it costs the established picture. Heavy-tailed interevent times are not generic once participation is chosen.

They survive only in a narrow window at the bifurcation and only at nonzero exploration, and there the exponent tracks the fraction of time spent coupled rather than the queue lengths, so $\alpha = 1 + 1/\max_j(L_j - 1)$ does not survive. We propose phase occupancy in its place; read the other way, exponents close to that family in correspondence data are evidence that the priority assignment there is close to exogenous.

On networks and in groups. Attention divides as $1/(k + a)$, fixing a critical degree k_c beyond which no coupled state exists, and the solitary phase percolates by Molloy–Reed with the second moment truncated at k_c — the intentional-attack calculation [16] with no attacker, which makes a scale-free contact structure maximally exposed.

Local structure decides how the failures are arranged rather than which relations fail. For activities requiring m participants the critical degree falls as the $(m - 1)$th root, giving a maximum group size that a reward proportional to m raises but cannot remove. Both the finite capacity for keeping ties active and the decay of ties whose rhythm is interrupted have been measured in communication records [11–14]; here they are k_c and the memory-timescale condition, derived rather than posited.

What is established and what is not. Exact and confirmed: the pair phase boundary and its validity condition. Measured and robust: the discontinuous, absorbing character of the transition; the price of recovery in units of τ_{mem}; the decay of relation survival with degree; and the fact that the coupled phase on a network is sustained by exploration rather than merely perturbed by it. Preliminary: the location of the network threshold, which still moves with run length and which the mean-field estimates overshoot by factors of 1.9 and 1.3; anything concerning critical behaviour, since we measure no exponent and attempt no scaling collapse; and the group results at $m \geq 3$, where the mean-field criterion fails qualitatively and what replaces it is a mechanism rather than a theory.

Provenance. Several results were reached while examining a different hypothesis, that experience is a property of systems whose preferred states are reachable only through another system they do not fully control; nothing in Secs. II–VII depends on it.

The commitment length is chosen from a fixed menu, the recovery constants use a memory time that agents attribute to their partners, and the network analogue of the partner-modelling agent is not implemented.

DATA AVAILABILITY

The data that support the findings of this article are openly available [36]. They are generated by the simulation code in that repository, which also contains the scripts that produce every table and figure.

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